Jennifer Chinyere

AI/ML Engineer | Backend Engineer | Data Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

AI/ML and Full-Stack Engineer with 2+ years building and deploying production-grade intelligent systems. Proven track record of developing scalable AI platforms that deliver measurable clinical and operational impact. Key achievements include building an AI-powered healthcare referral system **(MediFlow)** that reduced document processing time by **92%** and developing a computer vision tool **(OsteoDetect)** that improved radiologist efficiency by **2.1x**. Expert at owning end-to-end feature development in agile environments and bridging the gap between complex technical decisions and cross-functional team goals.

TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, HTML, CSS, PHP, Bash, SQL, C++
- Frameworks: FastAPI, Django, React, Next.js, SQLAlchemy, Pydantic, Tailwind CSS, TanStack Query, Zustand
- AI/ML: OpenAI API, Groq Llama, Whisper, LiteLLM, Ragas, ChromaDB
- Machine Learning & Data Science: Scikit-learn, MLflow, NumPy, Pandas, NLTK, spaCy, Jupyter Notebook
- Computer Vision: YOLOv8, OpenCV, Grad-CAM, Tesseract OCR, PyTorch
- Databases & Cloud: PostgreSQL, MongoDB, Redis, AWS (S3, EC2, Lambda), Render, Vercel, Railway, Supabase
- AI Developer Tools: Cursor, GitHub Copilot, Claude Code, ChatGPT, Gemini, Windsurf
- DevOps & Tools: Docker, Git, GitHub Actions, CI/CD, REST APIs, Linux, PowerShell, Postman, Jira, Slack, Zapier
- Testing & Security: Pytest, HTTPX, Jest, Cypress, React Testing Library, JWT, OAuth2, RBAC

PERSONAL PROJECTS

MediFlow - AI Healthcare Referral Management Platform

FastAPI, PostgreSQL, React, TypeScript, Docker, AWS, OpenAI API, Pydantic, Groq Llama 3.1, Whisper, Tesseract OCR, Scikit-learn(TF-IDF, Cosine Similarity), WebSockets

- Architected a multi-tenant healthcare referral platform with 50+ RESTful APIs using FastAPI, PostgreSQL, React, TypeScript, AWS, and Docker, streamlining referral intake and scalable clinical workflows across multiple facilities.
- Built an AI-powered document intelligence pipeline using Tesseract OCR, Whisper, and Groq Llama 3.1 to automate document extraction, speech transcription, LLM summarization, and AI-assisted referral reason classification, reducing processing time from 27 minutes to 2 minutes (92% OCR accuracy).
- Implemented duplicate patient detection using Scikit-learn and fuzzy matching techniques, alongside JWT authentication, RBAC, WebSockets, and DICOM-ready storage, contributing to a projected 92% reduction in referral handling time and approximately 360 clinical hours saved monthly per clinic.

OsteoDetect - AI-Powered Bone Fracture Detection System

Python, FastAPI, PyTorch, YOLOv8, OpenCV, Grad-CAM, HTML5, CSS3, Docker, Vercel, Railway

- Developed a computer vision application for automated X-ray fracture detection using YOLOv8, FastAPI, and OpenCV, with inference times of approximately 3 seconds per radiograph.
- Integrated Grad-CAM explainability visualizations and confidence scoring to improve clinical trust and interpretability (estimated 85% confidence boost).
- Built RESTful inference APIs and an end-to-end imaging workflow enabling clinicians to upload radiographs, receive AI-assisted predictions, and visualize fracture localization through an intuitive web interface.
- Optimized diagnostic workflows through AI-assisted triage, reducing manual review bottlenecks, saving an (estimated 40–50 minutes per radiologist per shift, and 82% user satisfaction during testing).

EDUCATION

Diploma in Computer Science & Engineering — 2024

Akirachix — CGPA: 9.06/10